HU351 Mathematical Economics and Econometrics (University Elective)

Time: 3Hours | Max Marks: 50

Note: Answer any five questions.

Assume suitable missing data, if any

1.a Explain step-wise how Econometrics is used for decision making 6

1.b True or False: 1x4

a. In a regression analysis, the values are fixed for the explanatory variable. T

b. Newton's law of gravity is an example for deterministic phenomena T

c. The assumption of multicollinearity means that there should be relationship among regressors. F

d. If a quantitative variable has 'm' categories, we can introduce only 'm+1" dummy variable. F

2.a What is Stochastic Variable? Discuss significance of the Stochastic Disturbance Term with suitable examples. 6

2.b Discuss relation between avarage Cost(AC) and marginal cost (MC) curves 4

i. when AC is declimimg

ii. When AC is minimum

iii. When AC is rising

3.a Discuss the assumptions underlying the method of Least Squares- the Classical Linear Regression Model. 6

3.b Fill up the blanks 1x4

i. The dependent variable in regression analysis

is assumed to be ---Stochastic Variable---.

ii. Variable such as grades in mathematics, results of horse race, degree of satisfaction at a restaurant are example of ---Ordinal---

iii. Under the least square procedure, larger the μ_i(cap)(in absolute terms), the larger the ---Eq. sum of residuals---

iv. In confidence interval estimation, α= 5%, this mean that this interval include the true $\beta$with probability of ---95%---.

4.a Discuss significance of Monto Carlo Experiments with suitable examples. 6

4,b In a duopoly, if the demand and cost function are given as $P= 80 -0.4(q_1+q_2)$ 4

$TC_1= 4q_1$; $TC_2=0.4q_2^2$

Find out q_1, q_2, P, π_1 π_2. Also prove the second order conditions for profit maximisation by both the duopolists. Whether a rise of either duopolist's output level will cause a reduction of other's optimum level?

P= 36
=2560
-720
~~20080~~
30

149

5.a Suppose you want to study the behaviour of sales of automobile over a number of years and suppose someone suggests you try the following models: 6

$Y_t = \beta_0+\beta_1 t$

$Y_t = \alpha_0+\alpha_1 t+\alpha_2 t^2$

Where Yt = sale at time t and t= time, measured in years. The first model postulates that sales is a linear function of time, whereas the second model states that it is a quadratic function of time:

(i) Discuss the properties of these models.
(ii) How would you decide between the two models?
(iii) In what situation will the quadratic model be useful?

5.b Discuss application of Differential Equation for Dynamic Multiplier. 4

6.a What do you mean by Dummy Variable? What caution need to be taken in use of Dummy Variable? Discuss with suitable example. 6

6.b Consider the following regression results

$$UNt = 207491 + 1.1507Dt - 1.5294Yt - 0.8511\ (DtVt)$$
$$T = (26.896) \quad (3.6288) \quad (-12.5552) \quad (-1.9819)$$
$$R^2 = 0.9128$$

Where UN = unemployment rate, %
V = job vacancy rate, %
D = 1, for the period beginning in 2018-IV
= 0 , for period before 2018-IV
T= Time measured in quarters

Note: In the fourth quarter of 2018, the (then) Labour Government liberalized the National Insurance Act by replacing the flat-rate system of short-term unemployment benefits by a mixed system of flat-rate and (previous) earning -related benefits, which increased the level of unemployment benefits

(i) What are your prior expectations about the relationship between the unemployment and vacancy rates?

(ii) Holding the job vacancy rate constant, what is the average unemployment rate in the period beginning in the fourth quarter of 2018 fourth quarter? How do you know?

7.a Discuss ANOVA model? How it is different from ANCOVA models? Discuss with suitable example

7.b What is Chow Test? Discuss its significance in Economtric Analysis.

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ion in Dummy Variables (1) If a
has 'm' categories, introduce
) dummy variables or else du
p. (2) Category for which no D
are, benchmark, control etc (3)
rcept value B1 represents the m
enchmark category. (4) Coeff attac